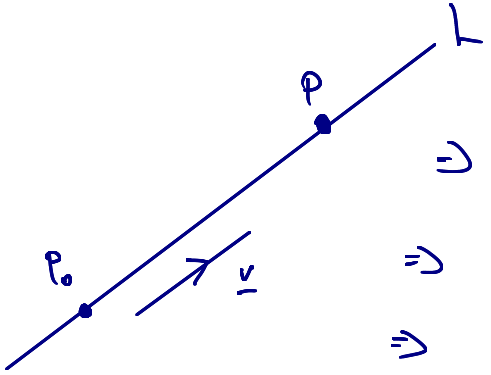
11.5 Lines and Planes in Space

To use scalar and vector products to write equations for lines, line segments, and planes in space.

Vector Equation for a Line

Suppose there is a line L through point $P_0(x_0, y_0, z_0)$ and it is parallel to the vector $\mathbf{v}$.

Let P be a point on L, then we know $\overrightarrow{P_0P}$ is parallel to $\mathbf{v}$.



$\vec{P_0P} = t\,\underline{v}$, $t \in \mathbb{R}$

$\Rightarrow \vec{OP} - \vec{OP_0} = t\,\underline{v}$

$\Rightarrow \vec{OP} = \vec{OP_0} + t\,\underline{v}$

$\Rightarrow \underline{r}(t) = \underline{r_0} + t\,\underline{v}$, $t \in \mathbb{R}$.

$\underline{r}(t)$ is the position vector of a point on the line L, and $\underline{r_0} = \vec{OP_0}$.

Parametric Equations for a Line

The standard parametrization of the line through $P_0(x_0, y_0, z_0)$ parallel to

$\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$ is

$$x = x_0 + tv_1,$$
$$y = y_0 + tv_2,$$
$$z = z_0 + tv_3,$$

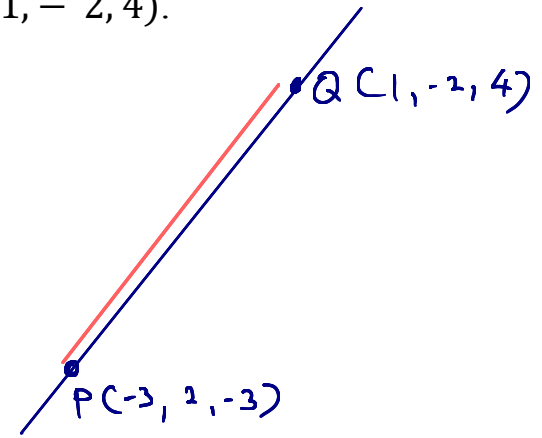
where $t \in \mathbb{R}$.

Example: What is the parametric equations for the line through the point $(-2, 0, 4)$ and parallel to the vector

$\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$?

$$\left.\begin{aligned} x &= -2 + tv_1 \\ y &= tv_2 \\ z &= 4 + tv_3 \end{aligned}\right\} \quad t \in \mathbb{R}.$$

Example: Find the parametric equations for the line through $P(-3, 2, -3)$ and $Q(1, -2, 4)$.



To describe the line segment PQ:

① $x = -3 + 4t$
$y = 2 - 4t$ $\quad 0 \le t \le 1$.
$z = -3 + 7t$

② $x = 1 + 4s$
$y = -2 - 4s$ $\quad -1 \le s \le 0$
$z = 4 + 7s$

L is parallel to $\vec{PQ} = \langle 4, -4, 7 \rangle$

Use P as our P_0:

$$\left.\begin{aligned} x &= -3 + 4t \\ y &= 2 - 4t \\ z &= -3 + 7t \end{aligned}\right\} \; t \in \mathbb{R}.$$

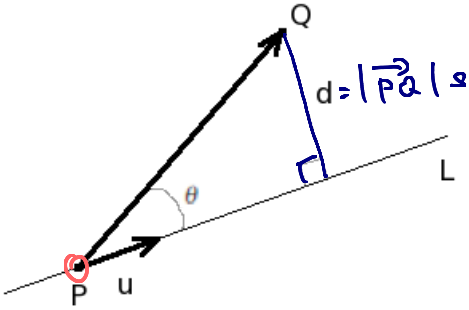
Use Q as our P_0:

$$\left.\begin{aligned} x &= 1 + 4s \\ y &= -2 - 4s \\ z &= 4 + 7s \end{aligned}\right\} \; s \in \mathbb{R}.$$

they describe the exact same line.

The Distance from a Point to a Line in Space

Suppose L is a line parallel to $\mathbf{u}$ and Q is a point not on L.



Let P be any point on the line L, then we have

$$d = |\overrightarrow{PQ}| \sin\theta$$

Recall that $|\vec{PQ} \times \underline{u}| = |\vec{PQ}||\underline{u}| \sin\theta$

$$\Rightarrow d = \frac{|\vec{PQ} \times \underline{u}|}{|\underline{u}|}$$

Example: Find the distance from point $S(1, 1, 5)$ to the line

$$L: x = 1 + t,\ y = 3 - t,\ z = 2t.$$

Choose $P = (1, 3, 0)$ [a point on L]

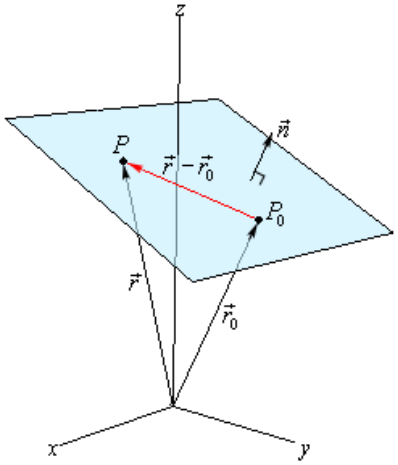
A vector parallel to L is $\underline{u} = \langle 1, -1, 2 \rangle$

$\vec{PS} = \langle 0, -2, 5 \rangle$

$$|\vec{PS} \times \underline{u}| = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 0 & -2 & 5 \\ 1 & -1 & 2 \end{vmatrix} = |\underline{i} + 5\underline{j} + 2\underline{k}| = \sqrt{30}$$

$|\underline{u}| = \sqrt{6}$ $\Rightarrow$ distance $= \dfrac{|\vec{PS} \times \underline{u}|}{|\underline{u}|} = \dfrac{\sqrt{30}}{\sqrt{6}} = \sqrt{5}$

An Equation for a Plane in Space



A plane in space is determined by a point in the plane and a vector perpendicular or normal to the plane.

Let $P_0(x_0, y_0, z_0)$ be a point in the plane M and let $\mathbf{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ be a vector normal to the plane M.

If $P(x, y, z)$ is a point in the plane, then the vector $\overrightarrow{P_0P}$ is perpendicular to $\mathbf{n}$, implying

$$\overrightarrow{P_0P} \cdot \mathbf{n} = 0.$$

$\Rightarrow (\vec{OP} - \vec{OP_0}) \cdot \underline{n} = 0$

$\langle x - x_0, y - y_0, z - z_0 \rangle \cdot \langle A, B, C \rangle = 0$

$\Rightarrow A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$

$\Rightarrow Ax + By + Cz = Ax_0 + By_0 + Cz_0$

Equation for a Plane

The plane passes through $P_0(x_0, y_0, z_0)$ and perpendicular to $\mathbf{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ can be represented by

- Vector equation: $\overrightarrow{P_0P} \cdot \mathbf{n} = 0.$
- Component equation: $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$

Usually written as $Ax + By + Cz = D$, where $P(x, y, z)$ is any point on the plane & $D = Ax_0 + By_0 + Cz_0$.

Example: The plane passing through $P_0(-3, 0, 7)$ perpendicular to $\mathbf{n} = 5\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ has component equation: $5x + 2y - z = 5(-3) + 2(0) + (-1)(7)$

$= -22$

Example:

Find an equation for the plane M through $A(0, 0, 1)$, $B(2, 0, 0)$, and $C(0, 3, 0)$.

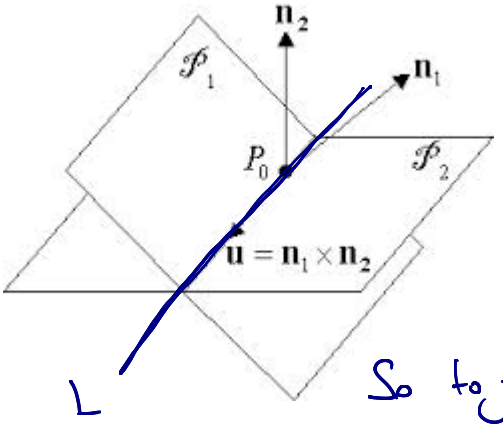
A vector normal to M is $\vec{AB} \times \vec{AC} = \langle 2, 0, -1 \rangle \times \langle 0, 3, -1 \rangle$

$$= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2 & 0 & -1 \\ 0 & 3 & -1 \end{vmatrix} = 3\underline{i} + 2\underline{j} + 6\underline{k}$$

Equation for M: $3x + 2y + 6z = 3(0) + 2(0) + 6(1) = 6$

Lines of Intersection

Just as lines are parallel if and only if they have the same direction, two planes are *parallel* if and only if their normals are parallel, or $\mathbf{n}_1 = k\mathbf{n}_2$ for some scalar k. Two planes that are not parallel intersect in a line.



Let L be the line of intersection of the planes P_1 and P_2.

Since L lies in both P_1 & P_2, L is orthogonal to both $\underline{n_1}$ & $\underline{n_2}$

$\Rightarrow$ L is parallel to $\underline{n_1} \times \underline{n_2}$

So to find the equation of the intersection line of two planes, find the cross product of their normal vectors & a point on both planes.

P_1 $\underline{h}$ $\underline{n}_1$

Example: Find the equation of the line of intersection of the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.

$\underline{n}_1 = \langle 3, -6, -2 \rangle$

P_2 $\underline{h}$ $\underline{n}_2$

$\underline{n}_2 = \langle 2, 1, -2 \rangle$

let L be the line of intersection of P_1 & P_2

$$\Rightarrow L \;//\; \underline{n}_1 \times \underline{n}_2 = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 3 & -6 & -2 \\ 2 & 1 & -2 \end{vmatrix} = 14\underline{i} + 2\underline{j} + 15\underline{k}$$

If $S(a, b, c)$ is a point on L, then S is on both P_1 & P_2.

So we have

$$3a - 6b - 2c = 15$$
$$2a + b - 2c = 5$$

Let $c = 0$:

① $3a - 6b = 15 \Rightarrow a - 2b = 5$

② $2a + b = 5$

$\Rightarrow a = 3$, $b = -1$

So the point $S(3, -1, 0)$ is on L.

$\therefore$ Equation of L :

$$\left.\begin{aligned} x &= 3 + 14t \\ y &= -1 + 2t \\ z &= 15t \end{aligned}\right\} t \in \mathbb{R}$$

Example: Find the point where the line

$$x = \frac{8}{3} + 2t,\ y = -2t,\ z = 1 + t$$

intersects the plane $3x + 2y + 6z = 6$.

Suppose $S(x, y, z)$ is the point of intersection.

$\Rightarrow \quad 3\left(\frac{8}{3} + 2t\right) + 2(-2t) + 6(1 + t) = 6$

$8 + 6t - 4t + 6 + 6t = 6$

$8t = -8$

$\Rightarrow t = -1$

$\Rightarrow \quad x = \frac{8}{3} + 2(-1) = \frac{2}{3}$

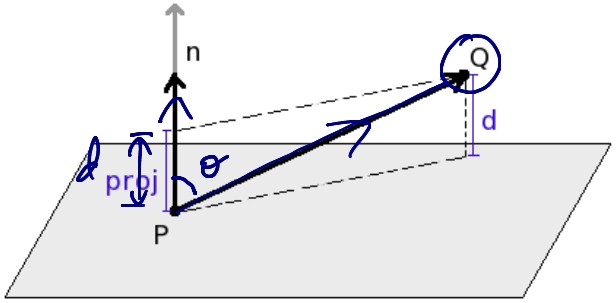
$y = 2$

$z = 0$

$\therefore$ the point S is $\left(\frac{2}{3}, 2, 0\right)$.

Recall : $|\vec{PQ} \cdot \underline{n}| = |\vec{PQ}|\,|\underline{n}| \cos\theta$.

The Distance from a point to a Plane



If P is a point (any point) on the plane with normal n, then the distance from any point Q to the plane is the length of the vector projection of $\vec{PQ}$ onto n. That is, the distance from Q to the plane is

$$d = \frac{|\overrightarrow{PQ} \cdot \mathbf{n}|}{|\mathbf{n}|}.$$

$d = |\vec{PQ}| \cos\theta$

$= \dfrac{|\vec{PQ} \cdot \underline{n}|}{|\underline{n}|}$

Example: Find the distance from $Q(1, 1, 3)$ to the plane $3x + 2y + 6z = 6$.

$P(0, 0, 1)$ is a point on the plane.

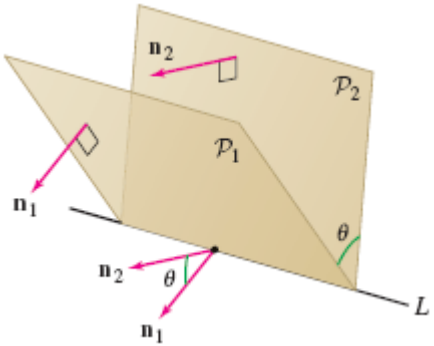
$\underline{n} = \langle 3, 2, 6 \rangle$

So the distance is $d = \dfrac{|\vec{PQ} \cdot \underline{n}|}{|\underline{n}|} = \dfrac{1 \cdot 3 + 1 \cdot 2 + 2 \cdot 6}{\sqrt{9+4+36}}$

$\vec{PQ} = \langle 1, 1, 2 \rangle$

$= \dfrac{17}{7}$ unit

Angle Between Planes



The angle between two planes can be determined by their normal vectors.

The angle between two planes is the same as the angle between their normal vectors.

Example: Find the angle between the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.

$\underline{n}_1 = \langle 3, -6, -2 \rangle$ $\quad$ $\underline{n}_2 = \langle 2, 1, -2 \rangle$

Recall

$\underline{n}_1 \cdot \underline{n}_2 = |\underline{n}_1||\underline{n}_2| \cos\theta$.

$$\Rightarrow \cos\theta = \frac{\underline{n}_1 \cdot \underline{n}_2}{|\underline{n}_1||\underline{n}_2|} = \frac{6 - 6 + 4}{7 \cdot 3} = \frac{4}{21}$$

$|\underline{n}_1| = \sqrt{9 + 36 + 4} = 7$

$|\underline{n}_2| = \sqrt{4 + 1 + 4} = 3$

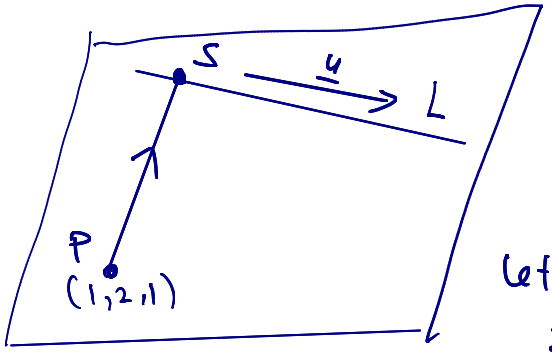
$\Rightarrow \cos\theta = \frac{4}{21} \Rightarrow \theta = \cos^{-1}\left(\frac{4}{21}\right)$

Example: Let M and N be the planes with plane equations:

$$M: 3x + y + 3z = 12 \qquad \vec{n}_M = \langle 3, 1, 3 \rangle$$

$$N: 2x - 3y + 5z = 19 \qquad \vec{n}_N = \langle 2, -3, 5 \rangle$$

Find the equation of the plane containing the line of intersection of the two planes M and N and the point $(1, 2, 1)$. Let L be the line of intersection of M & N.



L is parallel to $\langle 3, 1, 3 \rangle \times \langle 2, -3, 5 \rangle$

$$= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 3 & 1 & 3 \\ 2 & -3 & 5 \end{vmatrix} = 14\,\underline{i} - 9\,\underline{j} - 11\,\underline{k}$$

Let $S(a, b, c)$ be on L, so we have

$$3a + b + 3c = 12$$
$$2a - 3b + 5c = 19$$

$\Rightarrow$ L is parallel to $\langle 14, -9, -11 \rangle$ & contains the point $S(5, -3, 0)$.

Let $c = 0 \Rightarrow$ $\left.\begin{matrix} 3a + b = 12 \\ 2a - 3b = 19 \end{matrix}\right\} \Rightarrow \begin{matrix} a = 5 \\ b = -3 \end{matrix}$

$$\vec{PS} = \langle 4, -5, -1 \rangle$$

Now the plane we are interested in has normal

$$\vec{PS} \times \langle 14, -9, -11 \rangle$$

$$\vec{PS} = \langle 4, -5, -1 \rangle$$

$$\Rightarrow \underline{n} \text{ vector} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 4 & -5 & -1 \\ 14 & -9 & -11 \end{vmatrix}$$

$$= 46\underline{i} + 30\underline{j} + 34\underline{k}$$

Plane equation: $46x + 30y + 34z = 46(1) + 30(2) + 34(1)$

$$= 140$$

$$23x + 15y + 17z = 70$$